

The derivative of a composite function

Lessons 10–12 · Algebra 11, §1.5 · P1 · 7.5 · 28 problems

Differentiate with respect to x unless a rate is asked for. Factorise every answer — the factorised form is the one later questions need.

NAME _____

CLASS _____

DATE _____

Easy 7

1 $((x + 5)^3)'$

2 $((4x - 1)^2)'$

3 $((1 - 3x)^5)'$

4 $(\sqrt{5x})'$

5 $((x^3 + 2)^4)'$

6 $(\frac{1}{(3x + 1)^2})'$

7 $((2 - x^2)^6)'$

Medium 7

8 $((2x^2 + 3x)^5)'$

9 $(\sqrt[3]{x^2 + 1})'$

10 $(\frac{1}{\sqrt{4 - x}})'$

11 $((x + \frac{1}{x})^3)'$

12 $(x^2(3x - 1)^4)'$

13 $(\sqrt{x^2 - 4x})'$

14 The radius of a circle shrinks at 0.5 cm/s. Find $\frac{dA}{dt}$ when $r = 6$ cm.

Hard 7

15 Differentiate $y = (x - 1)^3(x + 2)^2$ and factorise the answer.

16 Differentiate $y = \frac{(2x + 3)^4}{x^2}$.

17 Differentiate $y = \sqrt{1 + \sqrt{x}}$.

18 Find the gradient of $y = (x^2 + 1)^3$ at $x = 1$.

19 Find the x -coordinates of the stationary points of $y = (x^2 - 9)^4$.

20 Differentiate $y = \frac{x}{\sqrt{x^2 + 1}}$.

21 The surface area of a cube grows at 12 cm²/s. How fast is the side growing when the side is 5 cm?

Very hard 7

22 Show that $(ax + b)^n$ differentiates to $an(ax + b)^{n-1}$ for every constant a , b and every index n .

23 Differentiate $y = (\frac{x + 1}{x - 1})^3$.

- 24 Find the equation of the tangent to $y = \sqrt{3x + 4}$ at $x = 4$.
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- 25 A spherical balloon is inflated at $100 \text{ cm}^3/\text{s}$. How fast is its radius growing when $r = 5 \text{ cm}$?
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- 26 $f(x) = (x^2 + k)^3$ has gradient 36 at $x = 1$. Find the possible values of k .
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- 27 Sand falls into a conical pile whose height always equals the radius of its base, at $8 \text{ cm}^3/\text{s}$. How fast is the height rising when $h = 4 \text{ cm}$?
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- 28 Show that $y = (x^2 - 2x + 2)^3$ has exactly one stationary point, and find it.
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